

Teaching Programming for Blinds: A Review

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Abstract— Many people do not have prior knowledge about what blinds can or cannot do as programmers. In fact, they can be efficiently competing with sighted peers when they have the required assistive tools. This paper provides a systematic review of published literature related to teaching programming for blinds between 1975 and 2013. Twenty-five reviewed papers from journals, conference proceedings, and scholarly databases formed the bases for this review. To the best of our knowledge, this is the first time to make a systematic review of literature relating to teaching programming for blinds. This review categorizes the literature based on different areas of interest, while providing quantitative analysis of publications based on their type and year of publication. Finally, future research venues are explored according to the gaps found in reviewed studies.

Keywords— *teaching programming; blinds; visually impaired; programmer; Audio Programming Language; Accessibility; Review*

I. INTRODUCTION

Universally there are nearly 285 million visually impaired people around the world: 39 million are blind and 246 are visually impaired, according to World Health Organization (WHO). Almost 90% of persons with low vision live in developing countries [1]. Vision is an essential sensory modality to get knowledge and learn. Traditional teaching techniques mainly rely on visualization to help in the understanding of complex information. There are many visual models used in teaching such as: diagrams, flowcharts, tables and images. Unfortunately, this type of teaching is not useful for visually-impaired students. However, there are some other methods which provide the information in non-visual manner but with less speed and ease of use compared with their counterparts.

Computing theory is considered a combination of logical and physical abstractions; in fact, computer science and information technology fields depend mainly on visualization in teaching different concepts. Specifically, learning programming is a key requirement for anyone who wishes to be a computer science student. However, it is a challenge for the majority of students with or without visual disabilities.

Despite significant increase of students' numbers in computer science majors, there are few visually impaired students who are enrolled in such majors, due to the additional efforts that are imposed on them, especially with the advancement of graphical user interfaces [2]. In spite of this,

computer programming is a chance for many blind and visually impaired persons to realize their creativity. Assistive technologies such as speech synthesis can help them to overcome many obstacles and limitations due to lack of vision.

This review paper explores the existing literature relating to teaching programming for blinds and classifies academic research based on a systematic review of the literature. Also it discusses some fundamental issues such as: the specific needs for blinds to learn programming, difficulties to novice blind learners to be integrated with conventional courses and if there is a need to develop specific programming languages for them.

II. METHOD

A. Data Collection

The main objective of this study is to investigate academic research related to teaching programming for blinds, and to give an understanding of the applied methods in the published literature and identify any shortcomings in the requirements.

Using methods that are similar to what is actually followed in other systematic reviews, a series of search efforts were performed to identify relevant papers. A paper was considered to be relevant if its main focus was on teaching programming for blinds or is concerned about blind learners in computer science major.

Firstly, some search terms were defined to be used without any limitation on the search period which is till the starting of this study (2013). Some used search terms were: Teaching programming for blinds, Programming and visually impaired and blind programmers.

Secondly, different academic databases were searched using the specified search terms. Databases include: ISI Web of Knowledge, IEEE Explorer, ACM, and Springer. Of course, a lot of unrelated papers were discarded. Google Scholar search engine was searched also with the same search terms. Again there were unrelated results.

Thirdly, the chaining technique in [3] was applied in order to get extra relevant papers. This resulted in more related papers in different databases. Also other publications which used the extracted papers as references were searched by Google Scholar (forward referencing).

B. Data Classification and Analysis

Both quantitative and qualitative classification was applied on the collected papers. Publication year and the type of

publication were used to perform the quantitative analysis, while an open coded content analysis was used to do the qualitative analysis. These classifications and analyses were implemented by reading papers more than once.

III. RESULTS

A. Quantitative Details

There is only little research related to the topic: Teaching programming to blinds. After filtering results based on the content, 25 papers were specified out of 46 found. However, Fig. 1 shows the distribution of published papers over years.

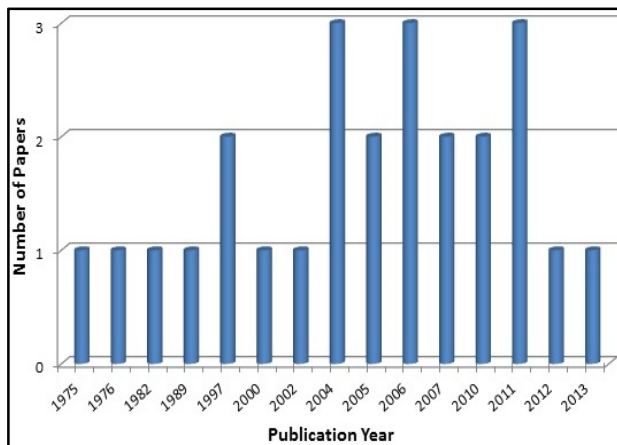


Fig. 1 Related Papers by Publication Year

Most of the identified papers appeared in conference proceedings (14 papers), with a lower number of journals articles (10 papers), and one book chapter. The percentage of papers is presented diagrammatically in Fig. 1 for each type of publication.

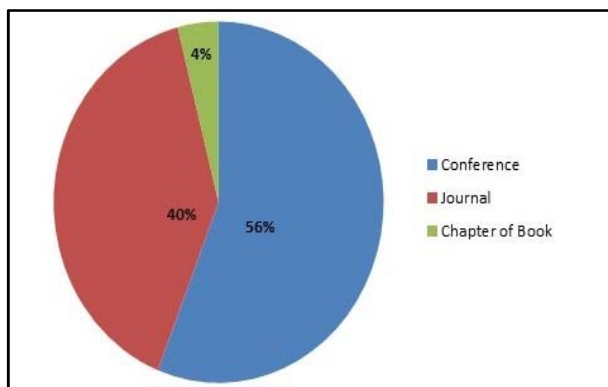


Fig. 2 Percentage of Papers Classified by Publication Type

Figure 3 shows the distribution of papers based on the year and publication type. From 1975 to 2013 there is a steady growth in the number of conference papers.

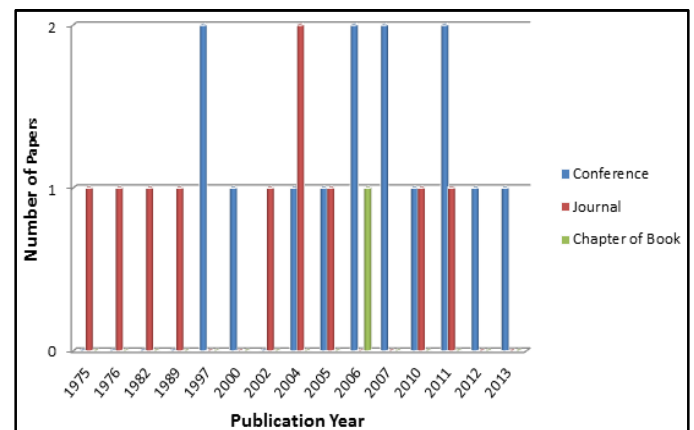


Fig. 3 Related Papers by Publication Year and Type

B. Publication Topics

Research papers in the literature can be classified into two different approaches: firstly, papers that are concerned about teaching programming for blinds e.g. [2, 4-17]. Even though some papers were not specifically for teaching programming for blinds, they were about teaching computing in general [11, 13, 14, 17]. Secondly, papers those make programming accessible for blinds. They assume that the blind already know how to program e.g. [18-24].

Another approach is to integrate blinds with sighted students in conventional courses. Some researchers discussed this approach by specifying special requirements, such as: providing the required Braille materials in order to prepare blind students for attendance in conventional courses and insuring that the lecturers have special skills for teaching blind students [4, 5, 12].

C. Programming Languages

The standard programming languages that appeared in the literature include: C++, Java, VB, C# and Assembly. On the other hand, some have developed new languages [9, 14, 16] such as: Audio Programming Language (APL), Human-Optimal Programming (HOP) and Graphical User Interface Description Language (GUIDL). Some studies focused on visual programming by blinds e.g. [16, 19-24]. It is worth mentioning that most proposed languages and tools were practically applied and evaluated, while some others were designed just as prototypes [16, 19].

APL, Audio Programming Language, was introduced to help blind grasp algorithmic thinking and problem solving. There is no concentration on code writing because the main objective is to learn programming concepts rather than syntax [15]. As a result, semantic errors of commands are avoided by ensuring that commands are correctly written [8]. However, an auto-completer helps in writing long names of classes or methods [10].

APL gives useful information regarding semantics and the program's organizational structure. The structure of the program is analysed and read to the blind in a similar way that separate lines and colours help sighted programmers. Also,

different sound emphases while audio reading help in rendering program text [2].

APL is not a new programming language; it is an audio based interface which interacts with blind programmers instead of the actual screen using text-to-speech systems to read the programming commands. It is compatible with screen readers such as JAWS [14]. It involves a new type of variables that stores and enables manipulating sounds, which are considered a fundamental unit by blind programmers [7]. The programs that are made by APL can be exported to Java in order to generate the corresponding Java code, then they can be used without the need of APL [9].

D. Assistive Devices

In addition to software tools such as text-to-speech and other audio-based techniques, which help blinds deal with computers, some special devices were used to facilitate learning computing, particularly programming for blinds. These devices include: Optacon, Braille writer as well as some manipulative tools which are used to explain concepts. Figures 4-6 present some of these assistive devices and tools.



Fig. 4 Optacon Scanner Device



Fig. 5 Braille Writer

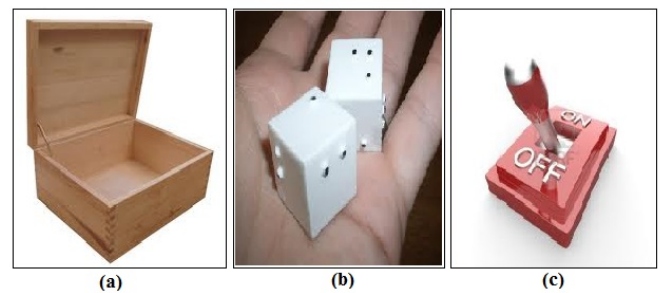


Fig. 6 Manipulative Tools for Teaching Programming Concepts, (a) Box represents variable, (b) Die represent numerical values, (c) Light switch represent Boolean values

IV. DISCUSSION

Computer programming was taught to blind students since 1960s. In 1998, there were 130 blind programmers according to the database of the American Federation for the Blind [24]. Nowadays, the issue of integrating blind with sighted students becomes more important. To perform the integration, two types of courses were discussed in [4]: lecture-based training and programmed instructions. In the former type, blind students should be supplied with Brailled manuals. While in the latter type, the text of programmed instructions is formed as audio books on cassettes. The in-house audio-visual education has helped blind programmers avoid the feeling of being disadvantaged [5].

When integrating blind students with sighted, it should be noted that blind students cannot read Braille with the same speed as sighted students read text. Furthermore, Braille content is usually not easy to navigate. Lack of awareness by trainers about the issues of vision impairment and training adaptation make the advancement and attempts to avoid difficulties more complicated. Because the required materials are still not ready, Coleman [4] proposed to setup a separate unit to rapidly produce Braille materials of all kinds.

Some programming languages were designed for both blind and sighted programmers. In such languages the sound was used for the blind programmers, while colours and fonts were used for sighted programmers [10]. However, there were some modes that require specific hardware such as: the pen mode which requires the use of tablet PCs [11].

Practically, designing an interface for a programming language is much harder than the design of an application interface [15, 19]. Through the design of programming languages for blinds, some issues should be considered, such as: compatibility with other languages and choice of representative words and symbols. Also it is not recommended to add new shortcut keys other than those familiar to the blinds. When using audio, small changes in word choice have large effect on the programmer understanding of the programming language. For example, the word “repeat” was chosen by blinds as the best representative word to introduce the concept of looping than other common used words such as: for, while, etc. [10, 13, 21, 22, 25].

Audio can be used as a sensory medium in order to help blind to interact while learning programming. Voice output by echoing each typed or displayed character is not useful due to

the resulted noise, especially like what happens with menu driven applications [26]. Audio-based languages were developed to help blind programmers write programs without the need to type or even memorize all instructions. They also help them to improve their ability for problem solving and thinking skills [7].

Generally, spoken commands should be said in a consistent way to limit the variation of vocabularies [2]. A good auditory cue should be short and "browsable" and begins with the most important information [14]. Some audio assistive tools uses their own sound cards while others use the sound card of the computer [27]. Stefik and Gellenbeck [13] claim that it is so difficult to design a language that works well with audio.

Some researchers have applied an efficient way to design their tools or languages. They asked blinds about their requirements by conducting preliminary interviews or surveys before starting the design in order to try to meet their requirements. Also feedbacks were collected from blind users about the actual design or by using a prototype. Many alterations were applied based on the notes and suggestions of experts and other users. Some languages, such as: APL, were designed by and for blinds.

V. LIMITATIONS

This study aimed present and classify existing literature relating to teaching programming for blinds. The study was based on a systematic consideration of the academic research. Unpublished papers and other resources are not included in this review because of credibility issues. Also published papers in languages other than English were not included.

It was observed that the vast majority of research focused on finding solutions to enable blind persons to use computing technologies. There is only little research dealing with teaching programming for the blinds.

VI. CONCLUSIONS

Although the main focus of most programming languages is on sighted people, computer programming plays a prominent role for visually impaired people as well. The use of screen readers that convert text into Braille or speech enables blind programmers to effectively work like their sighted peers.

This systematic study shows that teaching programming for blinds has a grown interest by researchers. Many articles published to date have discussed different perspectives regarding this issue. Most concepts in computer science are explained using examples with diagrams or animation before moving to the algorithm or code. However, teaching blind students require other ways that suits them. For example, blind students should work in groups and the tutor circulates around them instead of using the traditional teaching way, where the tutor stands at the front using a graphical screen.

In fact, blind learners depend on prior concrete experience, rather than building abstract thinking. So, they might not easily follow the same programming logic which is followed by sighted learners. However, the results of many interviews and surveys that were conducted by previous researchers have

clearly shown that blind programmers do not need new programming environments; instead they need simple and independent language to describe visual interfaces.

As discussed already, the compatibility issue should be taken into consideration when designing an assisting scripting language. Furthermore, it is better to separate the information and computations from the user interface to enable different ways for presentation such as: graphically, auditory, or any other way.

When concentrating on voice technologies, we should ensure that blind and visual impaired are not also hearing impaired. Pronunciation should be based on phonetic rules as well as dictionaries. Solutions need to integrate the hardware with the intelligent software in order to overcome some problems such as the pronunciation of words within different contexts.

On the other hand, the text-based interfaces of visual programming languages and their syntactical similarity with Visual Basic make them more promising for blind to enter the world of GUI (Graphical User Interface) programming.

In order to integrate blind students with sighted students, we should insure that blind students have received the same previous courses as sighted students. Lecturers, who are teaching courses that may be attended by blind students should have special skills even in talking, e.g. avoid using "this" for pointing, in order to avoid causing difficulties for blind students. Educating trainers of students with vision impairment has direct effect on their progress as well as self-confidence. Actually, the expertise in working with blind students does not necessarily insure the expertise in teaching them computer programming.

Through our analysis, we conclude that it is better to work directly with visually impaired students and programmers in order to know how exactly they want to learn programming. This is doable by employing techniques such as user centered design to build tailored programming languages/environments for the blinds.

Finally, a very important point is that blind and visually impaired persons are not necessarily the same. Also not all visually impaired persons are the same. May be it is more useful to produce tools for more than one category of users; this encourages a new area of research. The search for better alternatives of Braille is worthwhile, more than just providing formal computer education for blind students; this requires further research and exploration.

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